Catalog vs types: a clean-sheet choice

TYPED MESSAGES

- Define named structures for meaningful data groups.

- Consumers subscribe to those samples.

CATALOG-DRIVEN RECORDS

- Advertise data IDs, value types and meaning.

- Consumers choose points or coherent groups.

COMMON GROUND

- Both use DDS and typed values.

- Either can offer a catalog and a friendly selection UI.

Compare the models on their design merits, not on what has already been implemented.

The real choice is the selectable data unit

Question	Typed messages	Catalog-driven records	Design implication
Data identity	Named structure + instance	Data ID + instance	Both need stable identity
Value shape	Declared fields and relationships	Typed scalar or structured value	Neither must be untyped
Selectable unit	Message or declared group	Point or cataloged group	Choose granularity deliberately
Discovery	Types plus optional catalog	Catalog is a central contract	Catalog presence is not the whole choice

Typed samples can be small; cataloged values can be structured. Granularity is not predetermined.

Dictionary definitions are not status values

STATUS DICTIONARY

- navigation.speed: number, m/s, description.

- Defines schema, grouping and expected update behavior.

- Changes when the data definition changes.

STATUS VALUE

- navigation.speed = 3.2.

- Carries identity, timestamp and quality.

- Changes as the measured state changes.

WHY SEPARATE THEM?

- Avoid repeating static definitions in every update.

- Let users browse and select meaningful data.

- Define metadata and value bootstrap separately.

Receiving the dictionary does not supply the latest measurement values.

CFT selection: choose samples, not fields

TYPED SELECTION

- Select a message type/name or a small declared structure.

- A full navigation sample arrives as a unit.

CATALOG SELECTION

- Select data IDs or coherent groups.

- Independent points can be transmitted independently.

LIMIT SHARED BY BOTH

- CFTs do not extract one field from an accepted sample.

- Putting many points in one batch payload does not create field projection.

Point-level transmission requires point-level samples or explicit source-side extraction.

Structure and evolution: different places for the contract

Concern	Typed messages	Catalog-driven records	Shared responsibility
Validation	Named fields and type rules	Value kind + catalog/schema rules	Reject invalid values explicitly
Add a measurement	Add/evolve a compatible structure	Add a definition within the value vocabulary	Define meaning and constraints
Structured values	Relationships in the declared type	Relationships in a referenced schema	Version compatibility deliberately
Governance	Type definitions and semantics	Dictionary, value vocabulary and semantics	Assign an authoritative owner

A generic envelope can reduce type proliferation; it does not eliminate schema maintenance.

Consistency: preserve groups where they matter

STRUCTURED SAMPLE

- Related fields share one update.

- Natural fit for a coherent navigation or mission state.

- No implied transaction across separate writers.

INDEPENDENT POINTS

- Different values may arrive at different times.

- Natural fit for independently consumed telemetry.

- Group revision/snapshot rules are needed when combining them.

COMMON VALIDITY MODEL

- Preserve source time and quality.

- Distinguish unchanged-good, stale, missing and deleted.

- Use coherent groups in either design when needed.

Selectivity and consistency must be designed together.

Efficiency: selectivity versus per-sample overhead

GROUPED DATA

- Amortizes envelope and network overhead.

- Efficient when receivers need most related fields.

- Can send fields a receiver does not use.

SEPARATE POINTS

- Can avoid sending unwanted measurements.

- Useful when interests are sparse and different.

- Adds sample, identity and state-management overhead.

WHAT DETERMINES THE RESULT

- Selection fraction and update rates.

- Grouping, batching and quality metadata.

- Fan-out, repair traffic and catalog/bootstrap cost.

Neither model is inherently smaller or faster. Compare equal fidelity and delivery semantics.

Routing and operator selection can be shared

DATA INTERESTS

- Choose a type, data ID or coherent group.

- Expose those choices through a catalog UI in either model.

AUDIENCE INTERESTS

- Target individuals or teams.

- Preserve explicit global/control rules.

- Update CFT interests rather than team DDS partitions.

OBSERVED STATE

- One logical update per affected reader, not one guaranteed network message.

- Show desired, accepted and remotely observed interests.

Changing the value model does not require changing audience semantics.

Events and status have different design goals

EVENTS

- Reliable + TRANSIENT_LOCAL is a requirement.

- Commands retain structure, IDs and ACK/results.

- Set bounded history, expiry and replay rules.

STATUS VALUES

- Choose points or coherent state groups.

- Quality, freshness and latest-value behavior matter.

- Retention, bootstrap and batching remain open choices.

DICTIONARY

- Definitions must be available to new consumers.

- An unbatched reliable/TL profile is a candidate.

- Dictionary history and value history are different.

Transient local retains writer-held history; restart survival and safe command replay are separate.

A possible three-topic layout

EVENT

- Commands, ACK/results, alarms and other occurrences.

- Structured semantics and retained event identity.

- Transient-local durability as required.

STATUS

- Live values: points or coherent groups.

- Source/instance, timestamp and quality.

- Audience and data-selection metadata.

STATUS DICTIONARY

- Data IDs, schemas, units and descriptions.

- Group definitions and update expectations.

- Version, authority, availability and removal.

Shared discovery and routing do not require identical command and telemetry payloads.

Choose by the data, not by an incumbent

Primary need	Typed messages	Catalog-driven records	Clean-sheet conclusion
Coherent domain state	Natural declared structure	Use a cataloged structured group	Keep related values together
Independent telemetry	Use small measurement types	Reuse a generic point envelope	Choose the representation/governance fit
Frequent new points	Evolve types deliberately	Add definitions within the vocabulary	Catalog flexibility can be valuable
Commands and outcomes	Natural explicit request/result types	Require structured command schemas	Do not reduce control to an unchecked write

A mixed model is valid: structured events, selectable telemetry, and a catalog for both.

The remaining design decisions

DATA UNIT

- Which values are independent?

- Which must remain coherent groups?

- Does Event/Status really need one common type?

CATALOG CONTRACT

- Dedicated dictionary topic or another distribution path?

- Who owns schemas, revisions and removals?

- How do new consumers obtain definitions and values?

DELIVERY CONTRACT

- Event retention, expiry and safe replay.

- Status freshness, late join and restart recovery.

- Qualify batching/retained replay before relying on it.

Agree the semantic contract first; choose the wire representation and QoS to match it.

References: selection and resource semantics

[D1] Connext 7.7 CFT operations

https://community.rti.com/static/documentation/connext-dds/7.7.0/doc/manuals/connext_dds_professional/users_manual/users_manual/Using_a_ContentFilteredTopic.htm

Expression and parameter changes are separate supported operations.

[D2] Connext 7.7 SQL filter notation

https://community.rti.com/static/documentation/connext-dds/7.7.0/doc/manuals/connext_dds_professional/users_manual/users_manual/SQL_Filter_Expression_Notation.htm

WHERE-style sample selection; not field projection.

[D3] Connext 7.7 filtering resource policies

https://community.rti.com/static/documentation/connext-dds/7.7.0/doc/api/connext_dds/api_cpp2/group__content__filtering__qos.html

Filter state and property allocations need explicit sizing.

References: delivery, batching and value meaning

[D4] Connext 7.7 durability

https://community.rti.com/static/documentation/connext-dds/7.7.0/doc/manuals/connext_dds_professional/users_manual/users_manual/DURABILITY_QosPolicy.htm

Transient-local history depends on the writer remaining alive; restart recovery is separate.

[D5] Connext 7.7 batching

https://community.rti.com/static/documentation/connext-dds/7.7.0/doc/manuals/connext_dds_professional/users_manual/users_manual/BATCH_Qos.htm

Batching has history/resource and flush-delay considerations; no blanket batch/TL defect asserted.

[O1] SCADA reference: OPC UA DataValue

https://reference.opcfoundation.org/specs/OPC-10000-4/7.11

Value, quality/status, source and observation timestamps; no OPC UA interoperability claim.